INTISAR AHMAD

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OBJECTIVE

Final-year EEE student looking for a trainee engineer role in design verification or VLSI design. Hands-on with Verilog, VHDL and Cadence Genus, from RTL and testbenches to FPGA synthesis.

EDUCATION

B.Sc. in Electrical and Electronic Engineering (Major in Electronics) *Sept 2022 – Present*

BRAC University, Dhaka | Current CGPA: 3.68 | Expected CGPA: 3.70 | Expected Graduation: February 2027

Relevant Coursework:

VLSI Design, VLSI Design Laboratory, Digital Logic Design, Digital System Design, Embedded System Design, Computer Programming, Signals & Systems, Control Systems, Machine Learning, Modelling & Simulation

Higher Secondary Certificate (HSC) — Science *Jun 2019 – Dec 2021*

Dr. Mahbubur Rahman Mollah College, Dhaka | GPA: 5.00 / 5.00

Secondary School Certificate (SSC) *Completed Feb 2019*

IQRA Model School and College, Dhaka | GPA: 5.00 / 5.00

TECHNICAL SKILLS

Hardware Description Languages:

Verilog, SystemVerilog, VHDL (testbench design, FSMs, sequential circuits, pipelined datapaths)

EDA Tools:

Cadence Genus (RTL synthesis), Cadence Xcelium (simulation flow), Xilinx ISE (XST/ISim), GHDL, EasyEDA (PCB design)

Programming:

C, Python

Other:

MATLAB/Simulink, PCB Layout, MS Office Suite

PROJECTS

RISC-V RV32I 5-Stage Pipelined Processor (VHDL) *Xilinx ISE/ISim, Artix-7 FPGA*

- Built a 5-stage pipelined RV32I processor as part of a 5-member team, comparing EX-stage and ID-stage branch resolution
- Verified both variants (90 assertions, 5 testbenches); the ID-stage variant ran the test program in 59 cycles vs 67
- Synthesised on Artix-7 (xc7a100t): 1,613 LUTs at 158.6 MHz (ID-stage) vs 1,720 LUTs at 147.5 MHz (EX-stage)

Low-Power 16-bit ALU Research (VHDL/FPGA) *GHDL, Xilinx ISE, Artix-7 FPGA*

- Designed a 16-bit VHDL ALU with Radix-4 Booth multiply (8 cycles) and restoring divide (16 cycles) under a 4-state FSM
- Added an independent BIST FSM (self-test on reset), clock-enable power gating and shared Zero/Carry/Overflow/Parity flags
- Verified with 84 passing assertions across 6 testbenches (GHDL); synthesised on Artix-7: 690 LUTs, 199 registers

FPGA-Accelerated Point-of-Care Diagnostic Device *Research Paper — In Progress*

- Designing a quantized ML inference pipeline on FPGA for CBC blood panel analysis
- Developing parallel sensor-fusion modules and formal specification documents for subsystem verification
- Co-authoring a peer-reviewed research paper targeting international publication

VLSI Lab Projects — SISO Shift Register, FSM & Priority Encoder *Cadence Genus, Verilog*

- Synthesised a 16-bit SISO shift register in Cadence Genus; validated the gate-level netlist against RTL
- Wrote self-checking Verilog testbenches for an FSM and a priority encoder with corner-case stimulus

UAV/UGV Swarm Robotics — Final Year Design Project *MATLAB/Simulink, EasyEDA*

- Leading a 5-member team building a 4-robot swarm simulation for disaster response in Bangladesh
- Developed LoRa/ESP-NOW communication models; designed UAV circuit schematic for PCB fabrication

WORK EXPERIENCE

Mathematics and Science Teacher *Sept 2019 – Present*

IQRA Model School and College (IMSAC), Dhaka

- Teach Mathematics and Science to Classes 6–10
- Write lesson materials, set exams and mark scripts

Social Media Manager, Administrative Division *2022 – Present*

IQRA Model School and College (IMSAC), Dhaka

- Run the school's Facebook page

Founder, Wallchemy *2026 – Present*

Dhaka

- Online LED wall panel decor brand

EXTRACURRICULAR ACTIVITIES

- Senior Executive, Event Management — BRAC University Electrical and Electronics Club (BUEEC)
- Secretary of Operations — BRAC University Response Team (BURT)
- Editor & AD Manager — IQRA Model School and College (IMSAC)

LANGUAGES

English — Professional proficiency **Bangla** — Native